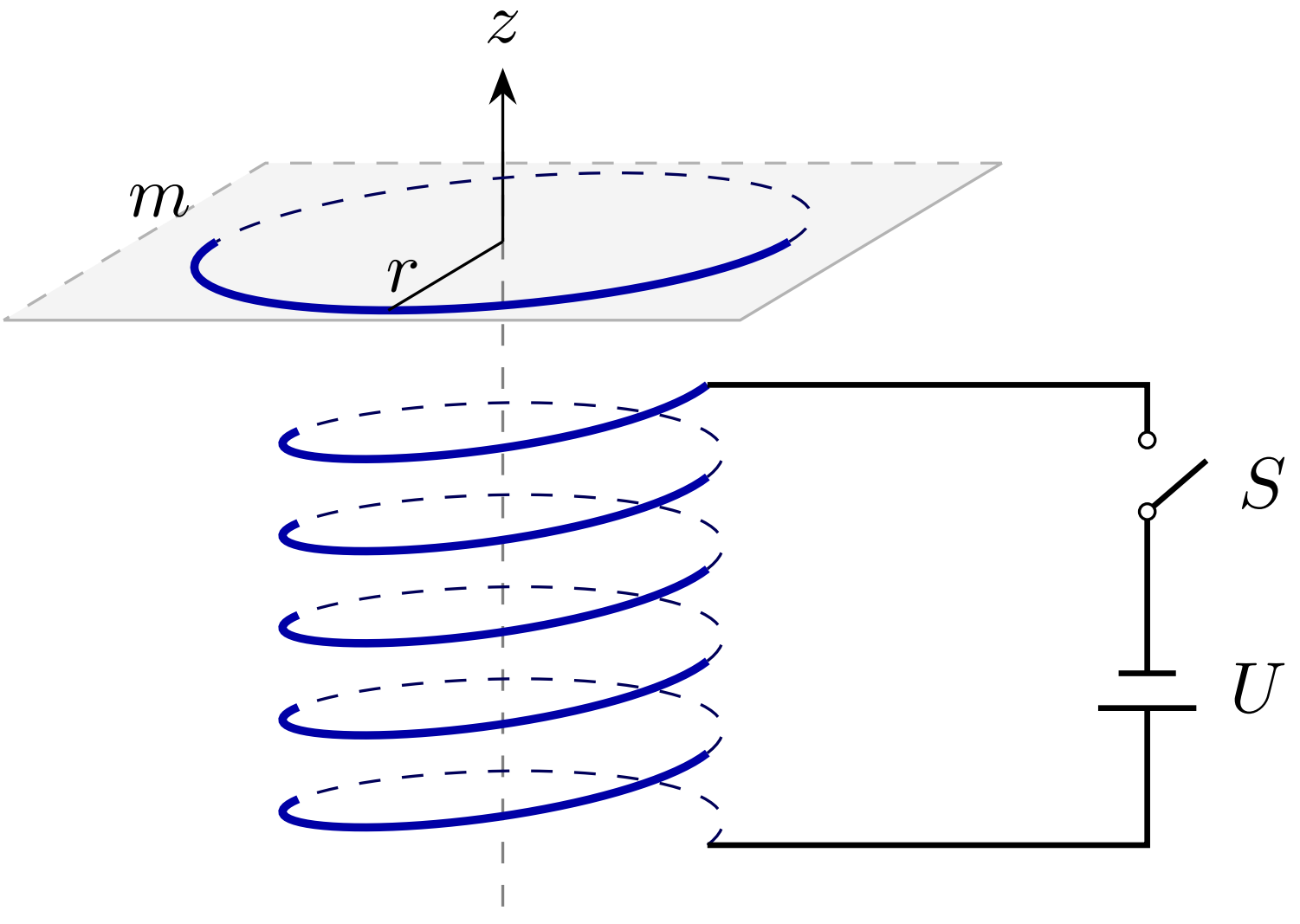
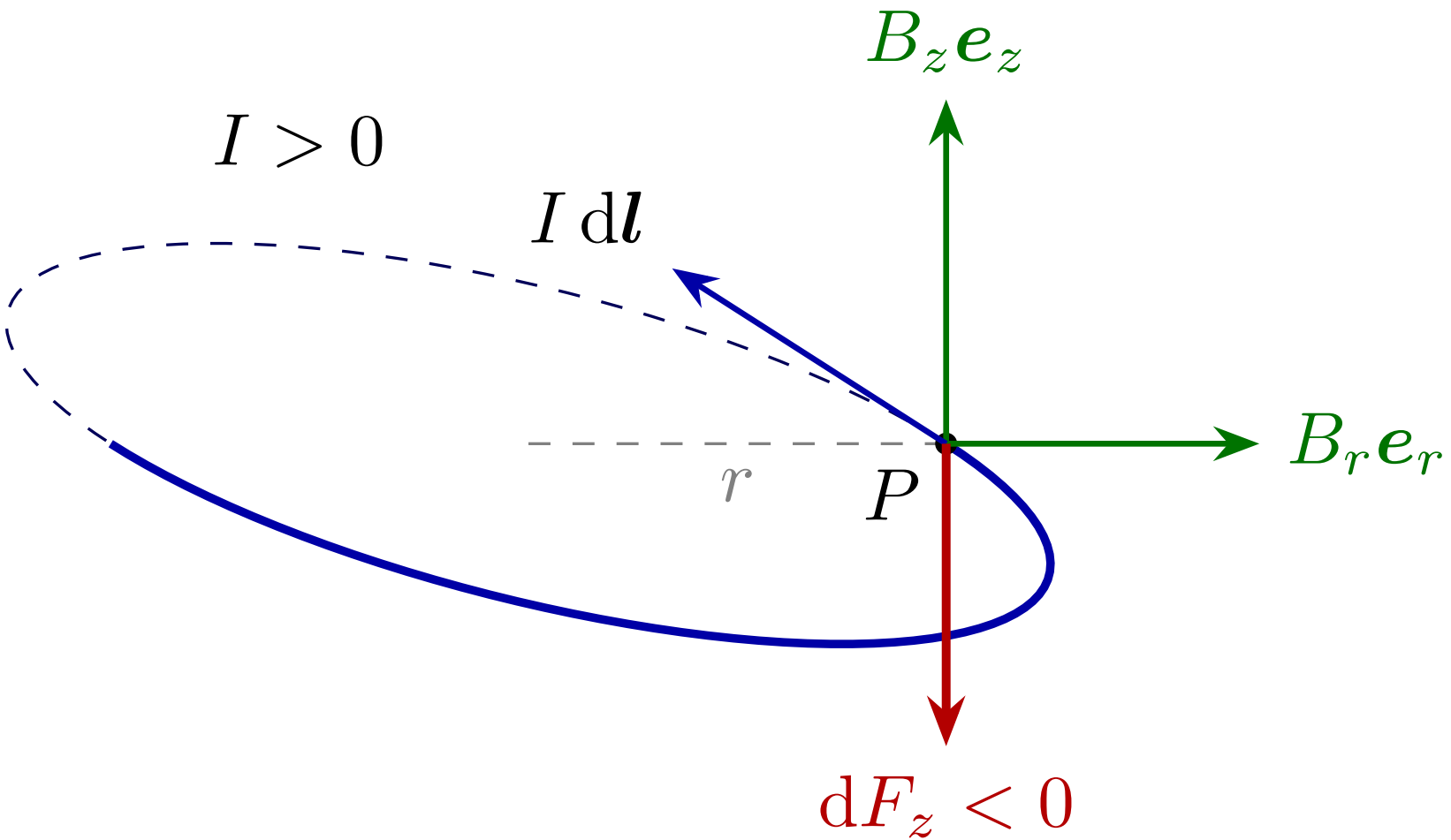
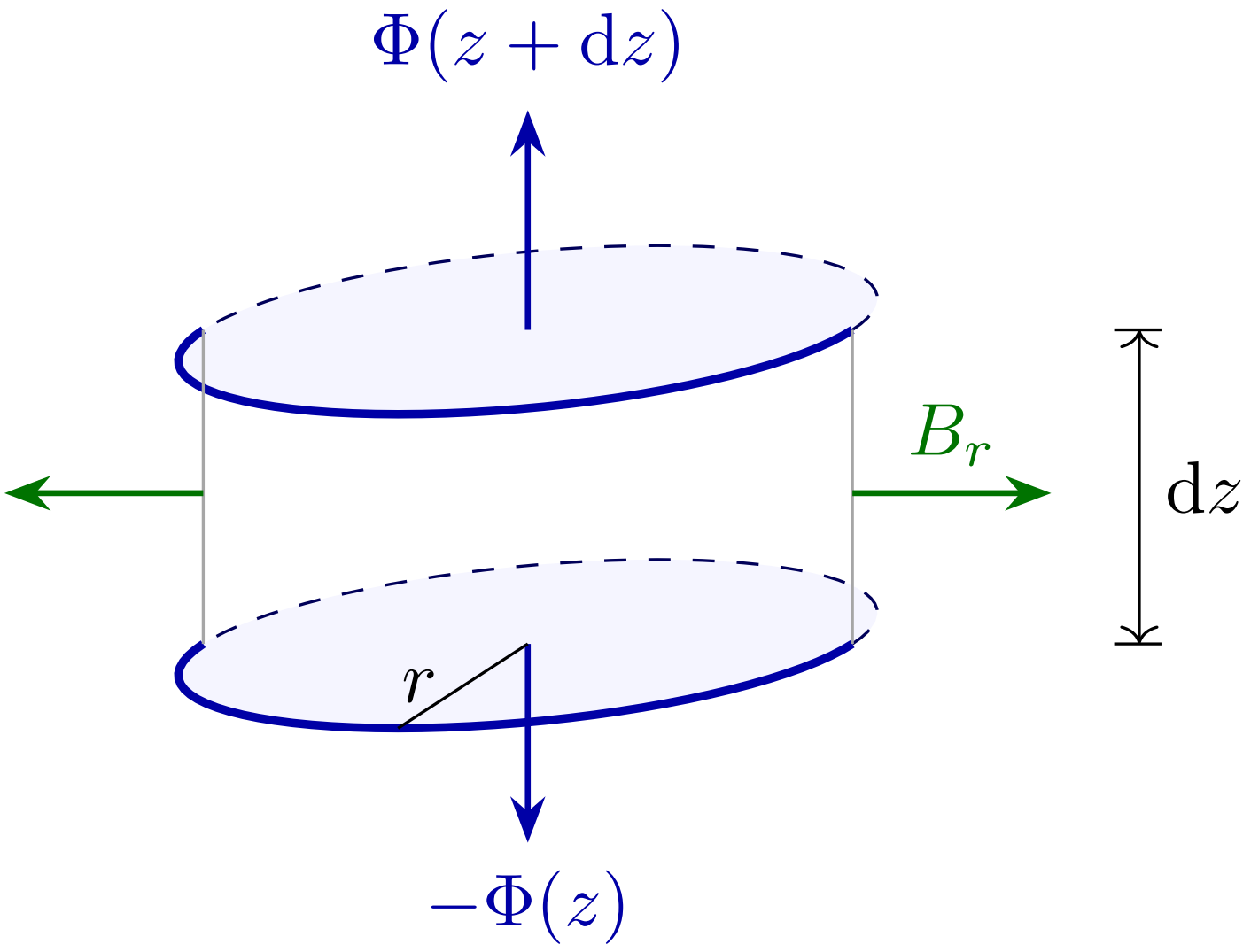
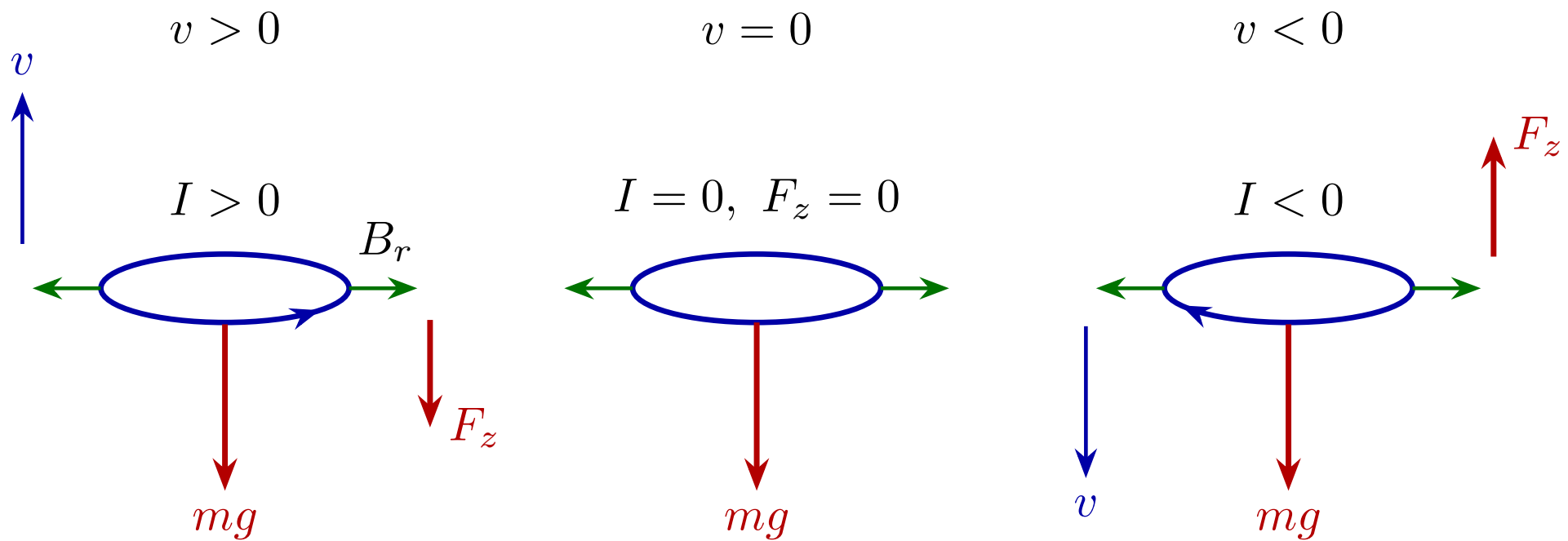
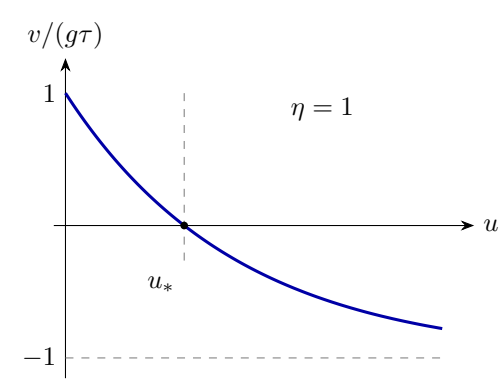
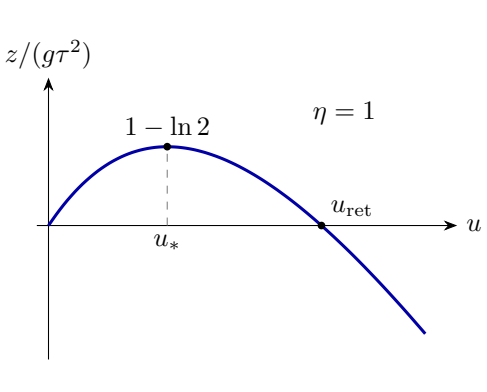

z
m
r
S
U


$B_z \boldsymbol{e}_z$
$I > 0$
$I\,\mathrm{d}\boldsymbol{l}$
$B_r \boldsymbol{e}_r$
r
P
$\mathrm{d}F_z < 0$


$\Phi(z + \mathrm{d}z)$
B_r
$\mathrm{d}z$
r
$-\Phi(z)$


$v > 0$
$v = 0$
$v < 0$
v
$I > 0$
B_r
F_z
mg
$I = 0,\ F_z = 0$
mg
$I < 0$
F_z
v
mg


$v/(g\tau)$
1
$\eta = 1$
u
u_*
-1


$z/(g\tau^2)$
$1 - \ln 2$
$\eta = 1$
u_{ret}
u
u_*